



University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and

Microcontrollers Lab

LAB REPORT

Experiment Number: 01

Experiment Title: Traffic Light Control using Servo Motor and LEDs

Submitted by:

Name : Md. Osman Goni

Student ID : 23101012

Section : A1

Submitted to:

Nakiba Nuren Rahman

Lecturer, Department of Computer Science and Engineering

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1. Experiment Name

Traffic Light Control using Servo Motor and LEDs

2. Objective

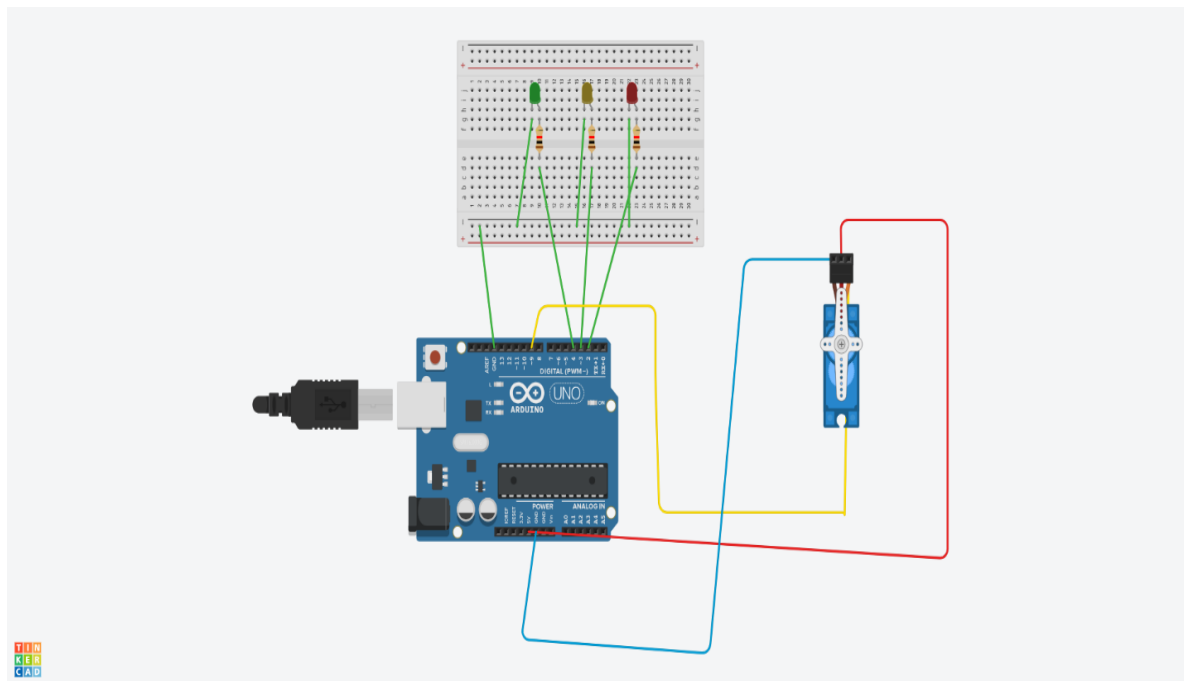
The objective of this experiment is to design and implement a traffic light control system using a microcontroller. LEDs are used to simulate red, yellow, and green signals, and a servo motor is used to represent a traffic barrier. The experiment helps to understand I/O interfacing, timing control, and PWM-based motor control in embedded systems.

3. Apparatus / Hardware & Software Requirements

- List all required tools and components:

- Microcontroller (Arduino, ESP8266 etc.)
- Sensors / Actuators (e.g., Ultrasonic Sensor, Buzzer, Relay)
- Software (e.g., Arduino IDE, EMU8086)
- Breadboard, Jumper Wires, Power Source etc.

4. Circuit Diagram / Schematic



5. Code / Assembly Program

```
#include <Servo.h>

Servo barrierServo;

int redLED = 2;
int yellowLED = 3;
int greenLED = 4;

void setup() {
  barrierServo.attach(9);
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
}

void loop() {

  // Green ON
  digitalWrite(greenLED, HIGH);
  digitalWrite(yellowLED, LOW);
  digitalWrite(redLED, LOW);
  barrierServo.write(90);
  delay(5000);

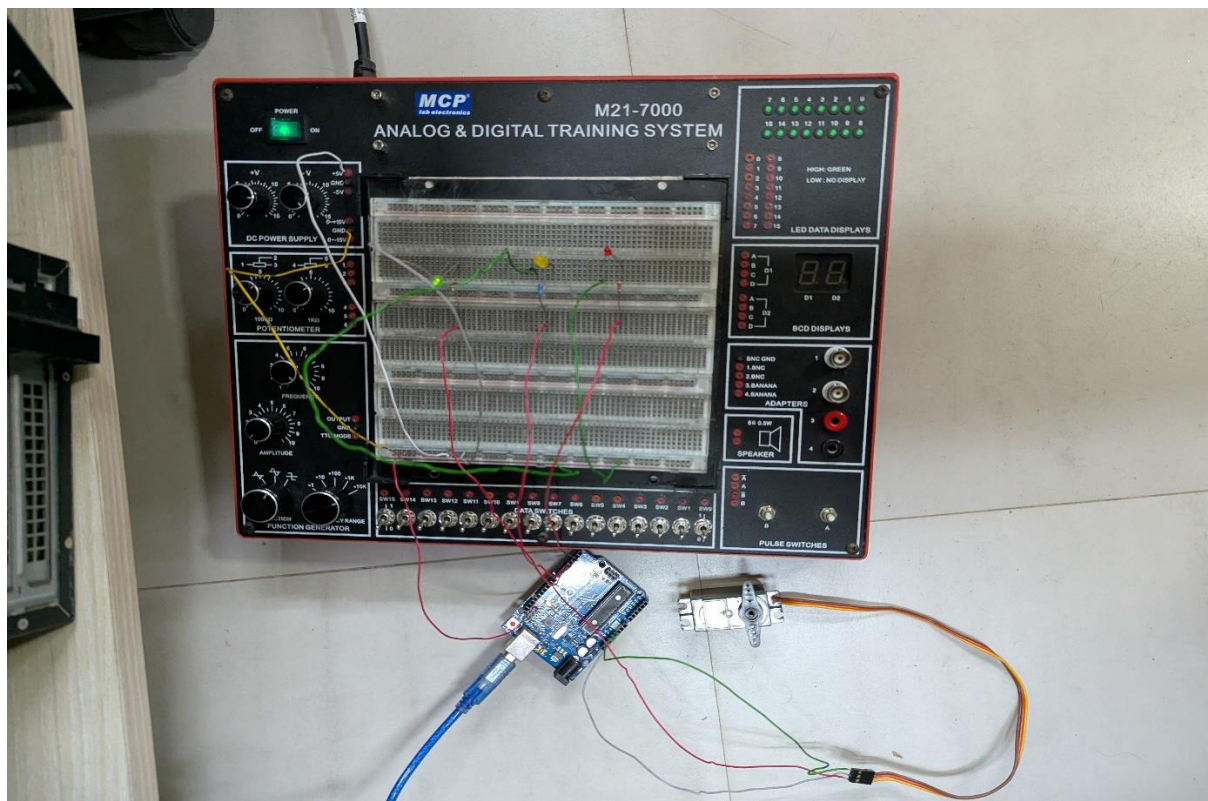
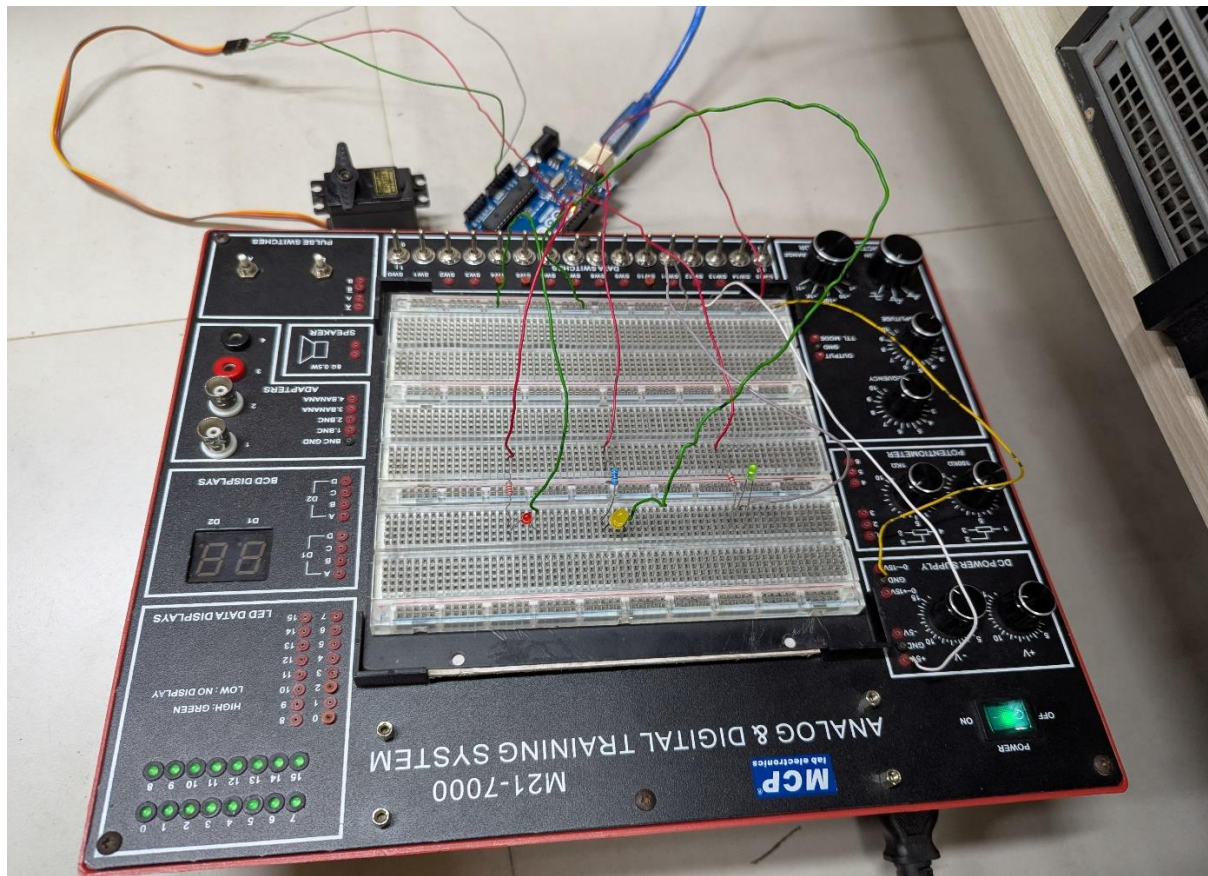
  // Yellow ON
  digitalWrite(greenLED, LOW);
  digitalWrite(yellowLED, HIGH);
  digitalWrite(redLED, LOW);
  delay(2000);
```

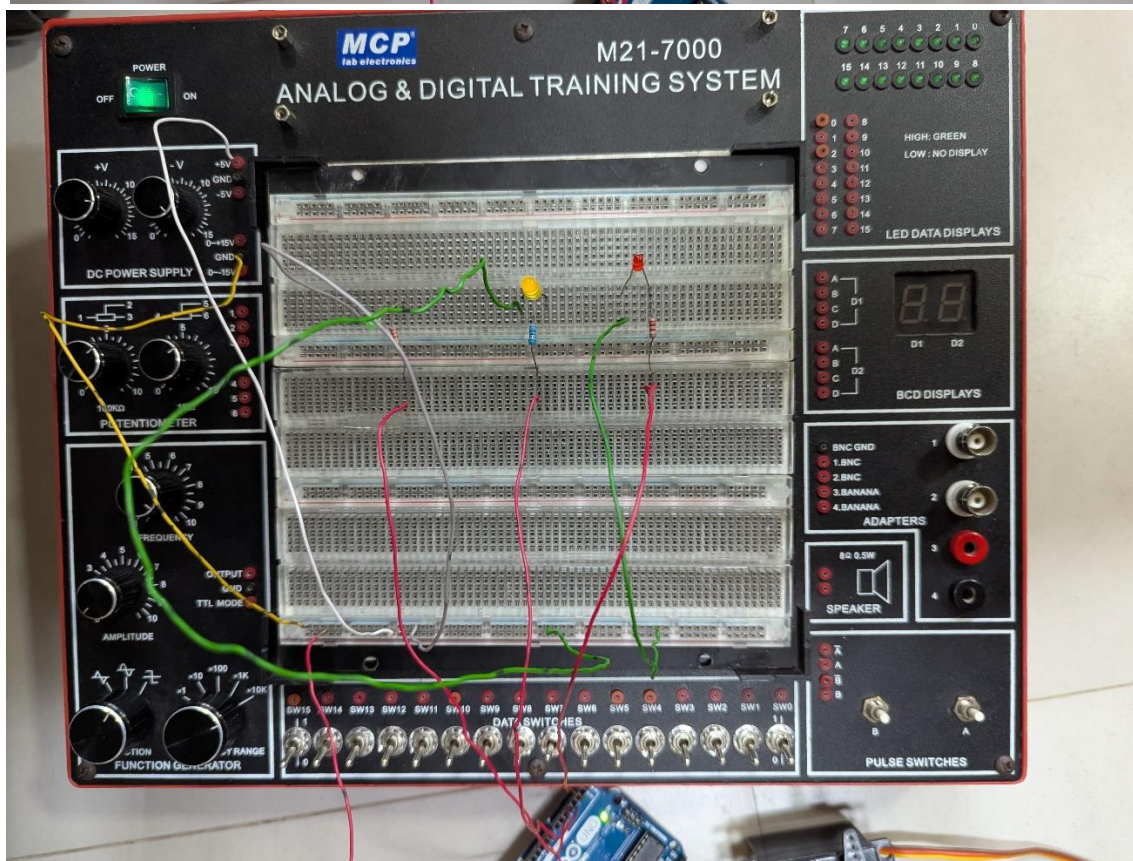
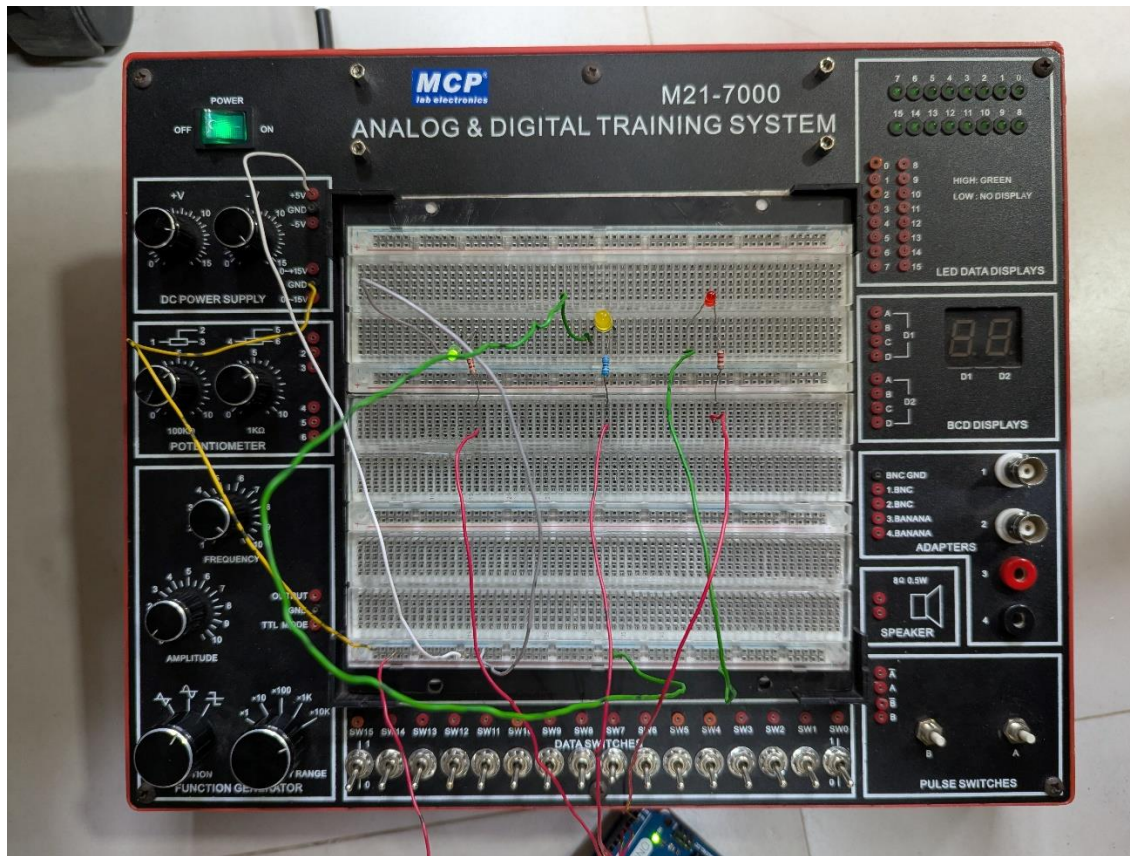
```
// Red ON  
digitalWrite(greenLED, LOW);  
digitalWrite(yellowLED, LOW);  
digitalWrite(redLED, HIGH);  
barrierServo.write(0);  
delay(5000);  
}
```

6. Output / Observations

The LEDs glow sequentially in the order of Green → Yellow → Red with a fixed time delay between each signal. The servo motor rotates to a specific angle to simulate the opening and closing of a traffic barrier (opens during green, closes during red). The yellow LED glows for a short transition period before changing signals. The complete cycle repeats automatically without manual intervention.

Proper PWM signals are generated to control the servo motor position accurately. The system operates smoothly without flickering or delay mismatch, confirming correct program execution and hardware interfacing.





7. Result

After implementing and testing the system, the following observations were recorded:

Green Signal Phase:

The green LED turned ON while the red and yellow LEDs remained OFF. The servo motor rotated to the open position, raising the barrier to allow movement. This phase operated smoothly and followed the programmed time duration accurately.

Yellow Signal Phase:

The system transitioned to the yellow LED after turning OFF the green LED, while the red LED stayed OFF. The servo motor remained in the open position during this short warning period. The timing and signal change occurred without interruption.

Red Signal Phase:

The red LED illuminated while the other LEDs were switched OFF. The servo motor moved to the closed position, lowering the barrier to stop movement. The system maintained this state for the specified time before restarting the cycle.

8. Conclusion

This project provided practical experience in working with microcontrollers and strengthened my understanding of embedded system concepts. By developing the traffic light control system, I was able to apply programming logic to real hardware components such as LEDs and a servo motor. The experiment improved my skills in writing structured Arduino code, controlling multiple output pins, managing servo angles, and maintaining proper timing coordination between different devices. It also enhanced my understanding of circuit connections, including correct pin configuration, resistor usage, and grounding techniques.

During the implementation process, I faced challenges such as ensuring correct wiring according to the program and adjusting delay timings to achieve smooth and realistic signal transitions. Solving these issues improved my debugging and problem-solving abilities. Overall, this project successfully connected theoretical knowledge with practical application and provided a solid foundation for future work in automation, embedded systems, and intelligent control projects.